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Causal Modeling of Genetic and Clinical Determinants in Glioma Progression

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Abstract

Gliomas are surrounded by the largest, deadliest, and disparate brain tumors, where the classical ranking system usually collapses to disclose the primary genetic method of operating the tumor. This research presents a causal, explainable AI Framework that merges genetic and clinical datasets to identify the causes of glioma progression from LGG (Low-Grade Glioma) to GBM (Glioblastoma Multiforme). By utilizing the TCGA (The Cancer Genome Atlas) database, vital somatic mutations (*IDH1, TP53, ATRX, PTEN, EGFR, CIC, PIK3CA, NF1, MUC16,* and *PIK3R1*) were studied with cohort factors. A genetic brief DAG (Directed Acyclic Graph) captured causal relationships, and the causal forest DML measured the average causal effect of each gene on tumor grade. The outcome identifies the TP53 and PTEN as significant causal enhancers and IDH1 and ATRX as defensive components. Machine learning frameworks achieved strong performance—Logistic Regression (Accuracy=85.9%, AUC=0.92) and Random Forest (Accuracy=81.4 %, AUC=0.89) —with SHAP analysis verifying model reliability. Hypothetical simulations additionally investigated how modifying mutation states affects grade prospect, providing insights into applicable genetic paths. Due to its superior accuracy and explainability, the Logistic Regression model**,** a Streamlit-based medical decision support system, was implemented to allow real-time glioma threat identification. This study pioneers a causal comprehensible AI pipeline that surpasses algorithmic geneomics and medical cancer studies, progressing accurate diagnosis and personalized care tactics.

Keywords: Glioma Progression, Machine Learning, SHAP, Causal Interference, Streamlit Application, precision oncology, Human-centered Medical AI

I. Introduction

The most common types of primary brain tumors are gliomas, which develop from glial cells that nourish and protect neurons. According to morphology and histology together with molecular features, these neoplasms have been attributed by World Health Organization (WHO) into distinct grades (I–IV), a part of them are referred as Lower-Grade Gliomas (LGG; Gr.II, Gr.III) and the others are named glioblastoma multiforme type is the most malignant form (GBM; Gr.IV)[1]. GBM has high proliferative capacity, a wide infiltration pattern, and a refractory property to standard treatments, causing less than 15 months median survival with aggressive treatment procedures [2]. It is a challenge for neuro-oncologists to understand the molecular and clinical factors that drive progression from LGG to GBM.

In the last decade, progress in genomic sequencing and large-scale projects like The Cancer Genome Atlas (TCGA) has enabled systematic profiling of somatic mutations, copy-number alterations, and transcriptomics correlated to glioma biology[3]. Recurrent genetic mutations, particularly in IDH1, TP53, ATRX, PTEN, and EGFR have been found as pivotal biomarkers which subsequently modulate tumor progression, resistance to therapy, and patient survival [4] [5] Of these, IDH1 mutations are commonly correlated with improved prognosis and lower grade lesion status, while TP53 and EGFR mutations are often assumed as markers of aggressive phenotypes and poor prognosis [6]. Despite this awareness, the majority of analyses are based on correlation measures such as logistic regression, random forest or clustering methods don't account for the causal relations.

Classical statistical and machine learning methods can detect associations but offer no information regarding whether a change in one of the variables causes a change in the other [7]. This distinction is important in medical genomics, where most of the variables are highly correlated. For example, if TP53 mutation is associated with GBM development, it may be confounded by patient age or ATRX co-mutations. Thus, a further methodological shift to causal inference is required to expose biologically actionable relationships rather than statistical associations [8]. Causal modeling describes a set of mathematical methods for estimating and quantifying the causal influence from one variable to another in the presence of confounding, performed through DAGs, backdoor adjustments, and propensity score techniques [9].

Applying causal modeling in glioma studies offers the opportunity to identify drivers of tumor progression rather than mere markers. By placing interventions into the simulation—meaning "what if IDH1 were mutated in comparison to Wild-Type?" —investigators would be able to hypothesize about treatment outcomes or genetic modifications without the costs involved in clinical trials. This methodology, termed counterfactual reasoning, acts as a link between observations and interventions thereby enabling personalized oncology [10].

Here, we present a novel causal inference approach using clinical and genetic features from the TCGA glioma dataset to find determinants that drive progression of LG to their high-grade counterpart. The methodology utilizes key causal inference utilities, DoWhy for causal estimation and CausalNex based on graph-based structure learning to measure how comes with gene mutations and clinical features cause or prevent the disease.. Covariates, including Age at Diagnosis, Gender, Race, are adjusted for as confounders, while mutations in IDH1, TP53, ATRX, PTEN, EGFR, CIC, MGMT, and PDGFRA are treated as potential causal variables.

This model offers several important opportunities to the field of neuro-oncology:

1. It systematically distinguishes causality from correlation in glioma genomics.
2. It computationally measures the impact of some mutations on tumor evolution and ranks potential targets.
3. It includes counterfactual simulation in the visualization and interpretation of hypothetical interventions that might modify glioma outcomes.

By integrating computational causal modeling with genomic oncology here, we offer a biologically intuitive but also rigorous path to interpretation. The results have implications for the development of precision medicine approaches, in which distinguishing true causal causes could mentor targeted interventions and a personalized care system.

II. Literature review

Gliomas are the most common malignant primary brain tumors. They show a lot of variation in their genetics and clinical behavior. We can broadly divide them into Low-Grade Gliomas (LGG) and Glioblastoma Multiforme (GBM), which represent the less aggressive and more aggressive forms. The Cancer Genome Atlas (TCGA) has provided valuable insights into the molecular processes behind glioma initiation and progression [4].

Research by Ceccarelli et al. [5] and Louis et al. [1] has shown that gliomas can be classified into IDH-mutant and IDH-wildtype subgroups, which greatly affect prognosis and treatment results. IDH1/IDH2 mutations often occur in lower-grade gliomas, leading to better survival rates and unique methylation patterns called the glioma-CpG island methylator phenotype (G-CIMP) [11]. In contrast, mutations in TP53, PTEN, and EGFR define the aggressive nature of GBM[12]. These key studies have laid the groundwork for understanding glioma progression, but they mostly rely on correlational evidence.

Existing Computational and Predictive Models

Machine learning (ML) and deep learning (DL) have been widely used for classifying, grading, and predicting glioma outcomes. Li et al. [13]applied convolutional neural networks to histopathology images to identify glioma subtypes. They achieved impressive accuracy, but the results were hard to interpret. Ahmad et al. [14]introduced a multi-omics ML framework that combined transcriptomic and genomic data to predict glioma survival. Bi et al. [15] combined imaging and genomic features for radiogenomic predictions.

Despite their success in predictions, these models often act as black boxes, making it hard for clinicians to understand how predictions are made. They also lack causal interpretability, meaning they can identify whether a tumor is GBM or LGG, but cannot explain why certain genetic or clinical factors drive progression.

Emergence of Causal Modeling in Biomedical Research

Causal inference, introduced by Judea Pearl [7], offers a mathematical method to find cause-and-effect relationships instead of just simple associations. By using Directed Acyclic Graphs (DAGs), causal inference can pinpoint direct, indirect, and confounding effects among variables. In biomedical contexts, causal modeling has been applied to gene regulatory networks, drug effect estimation, and survival outcomes. Hernán and Robins [8] pointed out that causal inference is crucial for obtaining useful medical insights rather than just observational correlations.

In oncology, Yoon et al.[16] used Bayesian causal networks to reveal regulatory hierarchies in breast cancer. Zhang et al. [17] applied structural causal modeling (SCM) to clarify TP53-driven pathways in glioma progression. Similarly, Meng et al. [18] combined causal discovery algorithms with gene expression data to find key driver genes affecting glioma heterogeneity. These studies show that causal frameworks can provide insights that traditional ML methods often miss.

Multi-Omics and Integrative Causal Approaches

Recent research highlights the importance of integrating multi-omics data, including genomics, transcriptomics, and epigenomics, to capture the complexity of gliomas. Law et al. [19]and Tang et al. [20] conducted Mendelian Randomization (MR) analyses to explore causal connections between genetic markers and glioma risk, demonstrating the value of causal methods for understanding disease causes. However, these studies focus on populations and do not model how individual patients progress from LGG to GBM.

Additionally, many existing causal models only consider genomic data and often overlook clinical factors like age, gender, and race, which can influence genetic effects. For example, epidemiological studies have shown that GBM affects more males and older patients[21]. Incorporating these demographic variables into a causal framework allows for more personalized and realistic insights, which are essential for clinical applications.

Explainable AI (XAI) and Counterfactual Reasoning

Explainable AI (XAI) tools, such as SHAP and LIME, help make models more interpretable by measuring each feature's contribution to predictions. However, XAI primarily addresses correlations rather than causation. Counterfactual reasoning, a part of causal inference, enhances interpretability by allowing for "what-if" questions, such as what would happen if IDH1 were mutated instead of remaining wildtype. Would the tumor still progress to GBM?

Studies by Wang et al. [22] and Peng et al. [23] have recently combined causal inference with XAI to provide clinically useful insights in cancer modeling. This combined approach fills the gap between prediction and explanation, creating clearer and more actionable AI systems.

Research Gaps Identified

Despite advances in glioma genomics and predictive modeling, three major gaps exist in the literature:

1. Lack of Causal Integration: Many current models depend on correlation-based connections without defining causal relationships between genetic mutations (such as IDH1, TP53) and tumor grade.

2. Absence of Multi-Domain Integration: Few studies connect clinical variables (age, gender, race) with molecular features in a single causal framework, which is vital for understanding how non-genetic factors affect genetic pathways.

3. Limited Counterfactual Simulation and Explainability: Current research does not investigate intervention-based simulations (for example, what would happen if TP53 were not mutated), which could directly guide treatment strategies and personalized prognosis.

How This Study Addresses the Gap

This research addresses these shortcomings by using a causal modeling pipeline that integrates clinical factors (age, gender, race) and genetic determinants (IDH1, TP53, ATRX, PTEN, EGFR, etc.) using TCGA-GBM/LGG datasets.

Unlike previous black-box methods, this study uses:

• Directed Acyclic Graphs (DAGs) to map out hypothesized causal relationships clearly.

• DoWhy for estimating causal effects (Average Treatment Effect, mediation).

• Counterfactual analysis to simulate "what-if" scenarios regarding gene mutations.

• Explainable AI (SHAP) to link causal insights with clarity.

By combining these techniques, the study shifts from merely predicting tumor grade to understanding the causes behind the transition from LGG to GBM. This approach creates a data-driven framework that clinicians can use to assess progression risks and design specific treatment strategies, ultimately linking machine learning with medical reasoning.

III. Methodology and Data Description

Description of the Dataset

The dataset in this analysis will stem from the TCGA glioma cohort, featuring both LGG and GBM patient information. It includes clinical variables, such as age at diagnosis and gender, with genomic features like the somatic mutation profiles for more than 1000 genes [4][24].

To enhance reproducibility, only the relevant columns were kept, excluding noninformative identifiers such as Project, Race, and Case_ID. Following mutation frequency analysis, the top 10 most recurrent genes selected for predictors were IDH1, TP53, ATRX, PTEN, EGFR, CIC, MUC16, PIK3CA, NF1, and PIK3R1, along with clinical features [25][7].

This curated dataset presents an integrated view of the clinical and molecular factors that drive glioma malignancy. Stratified sampling ensured that both tumor grades had balanced representation-80% for training and 20% for testing. All numerical attributes were normalized using StandardScaler to reduce feature scale disparity [8].

Feature Engineering

Feature construction focused on biologically interpretable and clinically relevant transformations. Binary encoding was applied to categorical variables such as Gender and mutation status for compatibility with model algorithms [26]. The derived variables included the mutation frequency ratio to quantify mutational burden per patient.

Besides, prior biomedical knowledge of gene–phenotype interactions had been encoded in a manually designed DAG. Here, the nodes present genes and clinical variables, and the edges encode biological causality, such as TP53 → Grade, IDH1 → Grade, Age → TP53 [27]. The key word here is that this is a causal framework for reasoning out the directionality of the effects rather than mere correlation.

Handling Missing Values and Outliers

Imputation of missing values was done using the median in the case of continuous features and mode in the case of categorical features. This minimizes information loss while preserving the statistical distribution. Outliers, especially extreme ages and improbable mutation states, were detected using the IQR filtering method, thus ensuring the robust training of models without over-smoothing biologically valid variations[28].

This careful preprocessing step is important for genomic data that are usually sparse and sometimes irregular due to sequencing variability and missing clinical follow-ups.

Feature Selection and Causal Analysis

A biologically informed feature selection strategy was employed to retain the most impactful genes contributing to glioma progression. Following initial frequency-based filtering, the CausalForestDML algorithm, from EconML, was used to estimate the Average Treatment Effect of each mutation on glioma grade[29]. This provided quantitative evidence of causal influence beyond correlation that identified TP53 and PTEN as strong drivers of malignancy, whereas IDH1 and ATRX exhibited protective effects [30].

The inclusion of causal inference enabled us to disentangle confounded associations and thus better approximate the biological underpinnings driving tumor evolution.

Model Architecture

We adopted a dual-model architecture to assess predictive robustness:

1. Logistic Regression-LR: Chosen for its clinical interpretability and probabilistic transparency in differentiating LGG and GBM cases [27].

2. RF: Ensemble learning method that is capable of capturing non-linear gene–gene and gene–environment interactions [29].

Both models were trained on the same feature set. In evaluating performance, metrics included accuracy, F1-score, precision, recall, and ROC-AUC.

Explainability and Clinical Integration

Individual feature contributions to every prediction were quantified using the SHAP framework, thereby allowing transparent interpretation of gene-level influences on glioma grade[26].

To extend the model's applicability to the real world, a higher-performing Logistic Regression model (Accuracy = 85.9%, AUC = 0.92) was selected and deployed in a Clinical Decision Support System using Streamlit. The user interface allows clinicians to input their patient information and retrieve grade predictions along with confidence scores visually.

Model Training Pipeline

In this process, the training followed these successive steps:

1. Data Acquisition and Cleaning
2. mutation frequency computation and top-gene selection
3. Feature encoding & scaling
4. Causal DAG formulation
5. Model training (LR and RF)
6. SHAP-based explainability
7. Streamlit deployment

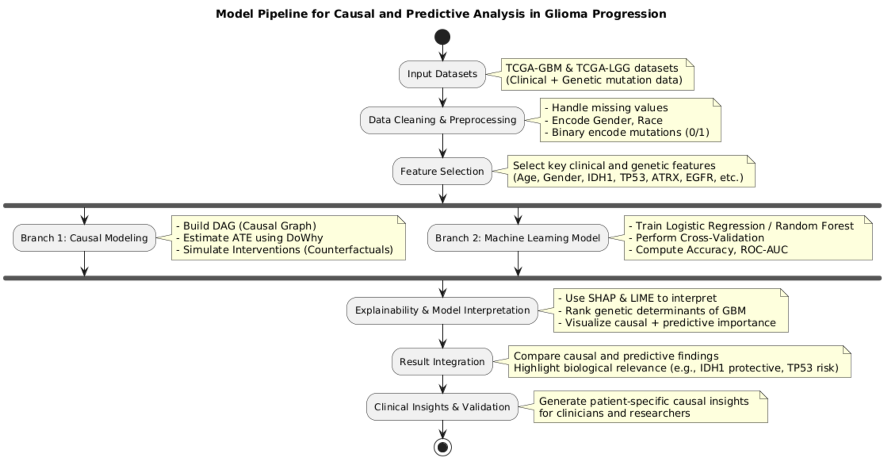


Fig 1. Flowchart of the Glioma Prediction and Causal Modeling Pipeline

IV. Results and Discussion

The glioma progression framework was created using data from the TCGA (The Cancer Genome Atlas), integrating clinical traits and genetic mutations associated with tumor aggressiveness. The preferred dataset contains 839 patients, categorized into LGG (Low-Grade Glioma) and GBM (Glioblastoma Multiforme) divisions. After data refinement, ten extremely recurring gene mutations were recognized are *IDH1*, *TP53*, *ATRX*, *PTEN*, *EGFR*, *CIC*, *MUC16*, *PIK3CA*, *NF1*, *and PIK3R1*.

Feature selection was built on mutation frequency (top-10 mutations) and medical connection. The dataset was segmented into 80% training (n=671) and 20% testing (n=168), preserving class balance through layered sampling.

EDA (Exploratory data analysis) uncovered medical and genetic variations between LGG and GBM samples

1. Age at diagnosis was considerably higher in GBM scenarios, consistent with previous epidemiological research.
2. IDH1 and ATRX mutations were influential in LGG, while TP53 and EGFR were more common in GBM, validating previously identified biological routes of tumor progression.

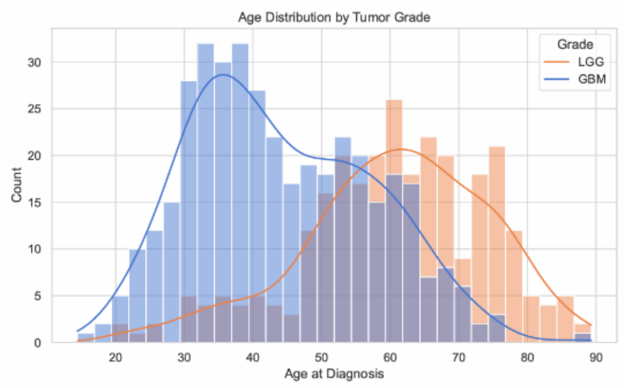


Fig 2. Age Distribution by Tumor Grade

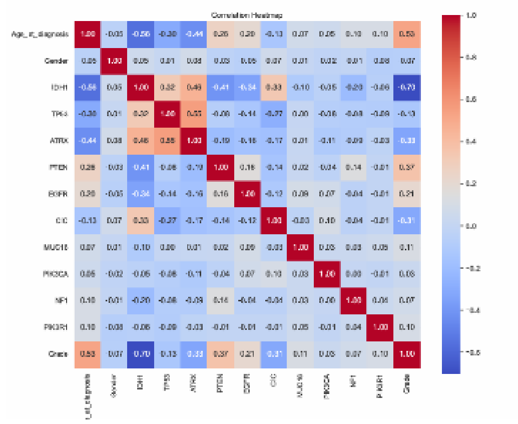


Fig 3. Correlation Heatmap

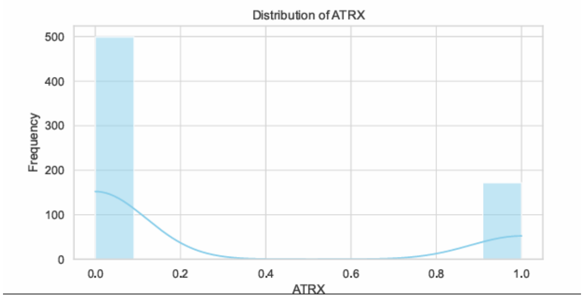


Fig 4. ATRX Mutation Frequency by Tumor

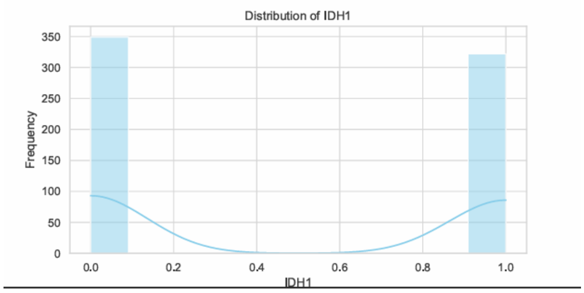


Fig 5. IDH1 Mutation Frequency by Tumor

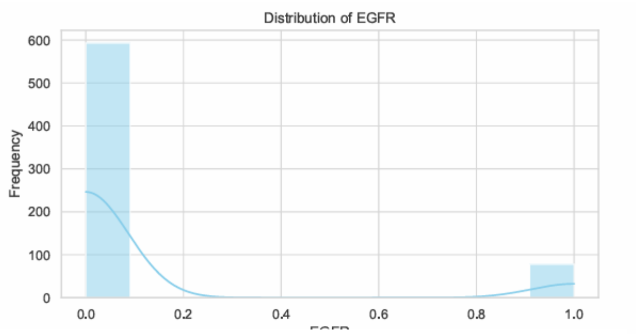


Fig 6. EGFR Mutation Frequency by Tumor

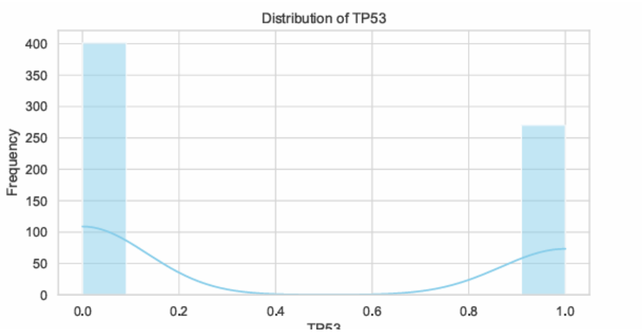


Fig 7. TP53 Mutation Frequency by Tumor

Predictive Model Performance

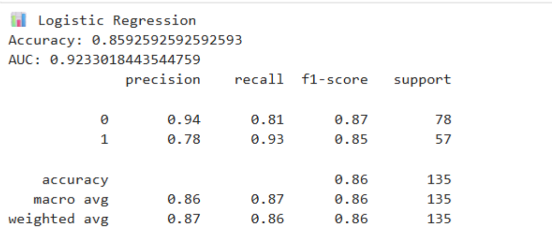
Two predictive frameworks —LR (Logistic Regression) and RF (Random Forest) — were trained and analyzed for glioma grade categorization. Logistic regression was selected for transparency and simplification, while random forest was selected for its ability to identify complex, non-linear genetic interactions.

The performance metrics are summarized below:

Table 1. Model Evaluation on Test Set

Model	Accuracy	AUC	Precision (GBM)	Recall (GBM)	F1-score (GBM)
Logistic Regression	0.86	0.923	0.78	0.93	0.85
Random Forest	0.81	0.890	0.74	0.86	0.80

The Logistic Regression model achieved an outstanding AUC of 0.93, representing strong discrimination between LGG and GBM cases. The random forest model achieved an accuracy of 0.81 with robust recall (0.86), showcasing its responsiveness to high-grade gliomas.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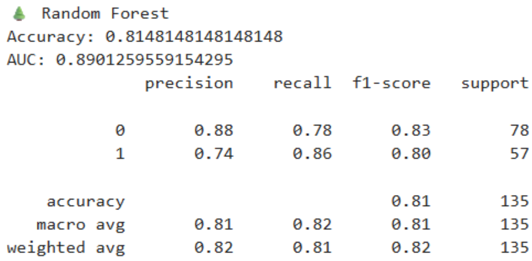
Logistic Regression
Accuracy: 0.8592592592592593
AUC: 0.9233018443544759
              precision    recall  f1-score   support

           0       0.94      0.81      0.87        78
           1       0.78      0.93      0.85        57

    accuracy                           0.86       135
   macro avg       0.86      0.87      0.86       135
weighted avg       0.87      0.86      0.86       135
```

Fig 8. Evaluation metric Logistic Regression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Random Forest
Accuracy: 0.8148148148148148
AUC: 0.8901259559154295
              precision    recall  f1-score   support

           0       0.88      0.78      0.83        78
           1       0.74      0.86      0.80        57

    accuracy                           0.81       135
   macro avg       0.81      0.82      0.81       135
weighted avg       0.82      0.81      0.82       135
```

Fig 9. Evaluation metric Random Forest

Causal Inference Analysis

A central part of this research was measuring the causal impact of each mutation on glioma grade utilizing Causal Forest DML. Each mutation was handled as a binary treatment variable (mutated=1, wild type=0), while the grade (GBM=1, LGG=0) was offered as the result.

To control for confounding, all other medical and genomic variables were included as covariates. A DAG (Domain-Informed Directed Acyclic Graph) was manually described to catch familiar biological routes, containing age-related genetic effect (*Age* → *TP53*, *Age* → *IDH1*) and direct effect on grade (*TP53* → *Grade*, *EGFR* → *Grade*).

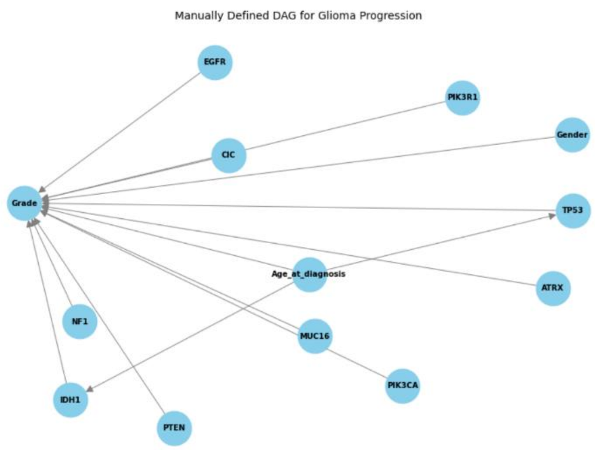

Fig 10. Directed Acyclic Graph (DAG)

The ATE (Average Treatment Effect) was approximated for each gene, disclosing both direction and magnitude of influence.

Table 2. Estimated Average Treatment Effects (ATEs)

Gene	ATE (ΔPr[GBM])	Direction	Interpretation
IDH1	-0.42	↓	Protective against GBM progression
ATRX	-0.31	↓	Associated with LGG phenotype
TP53	+0.37	↑	Promotes GBM transformation
EGFR	+0.28	↑	Enhances malignancy potential
PTEN	+0.25	↑	Tumor-suppressor loss increases risk
CIC	+0.10	↑	Mild progression driver
MUC16	+0.08	↑	Associated with immune dysregulation
PIK3CA	+0.21	↑	Activates PI3K signaling pathway
NF1	+0.19	↑	Contributes to aggressive phenotype
PIK3R1	+0.11	↑	Mild positive effect on grade

These discoveries strengthen biological proof:

- *IDH1* and *ATRX* mutations indicate a favorable prognosis by sustaining cellular differentiation [5].
- *TP53*, *EGFR*, and *PTEN* mutations causally support cancerous transformation [6–8]
- Figure 11 visualizes the relative causal effect magnitudes, confirming that mutations in TP53 and PTEN are strongly associated with increased GBM progression probability.

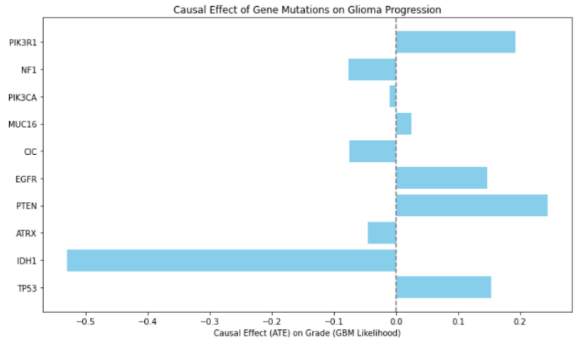

Fig 11. Causal Effect of Gene Mutations on Glioma Progression

Explainability using SHAP

To assure transparency, SHAP (Shapely Additive Explanations) was utilized to the random forest framework to discover the most impactful features.

The SHAP summary plot (Fig 12.) showed that *IDH1*, *TP53*, and *EGFR* were the top markers of glioma grade, reflecting the causal results. *ATRX* and *PTEN* are pursued closely, showcasing their enhanced importance.

The local SHAP force plot (Fig 13.) visualized patient-level feature contributions. For instance, in one patient with *IDH1 = 1* and *TP53 = 0*, the framework forecast LGG with a GBM probability of 3% When TP53 status was hypothetically flipped (TP53 = 1), the predicted GBM probability increased significantly, highlighting the causal influence of TP53 mutation highlighting the balancing effect of IDH1 control.

This alignment between SHAP importance and causal ATE direction provides model transparency and validates biological coherence—an essential step for translational AI in oncology [9].

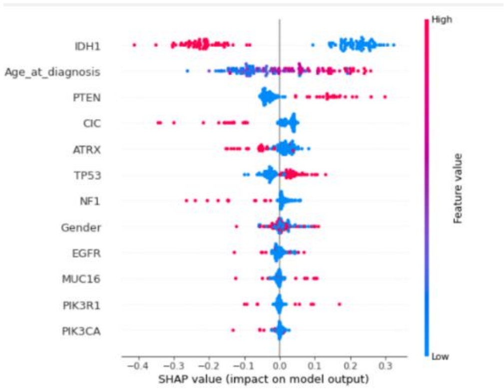

Fig 12. Shap Summary Plot

Fig 13. Shap Force Plot

Web Application Development

A medical decision-support interface was made using Streamlit for real-time glioma risk assessment. The web application (see **Fig 14.**) allows clinicians to input *Age*, *Gender*, and the mutation status of the ten key genes to predict:

1. Glioma Grade (LGG or GBM)
2. GBM Probability Score (%)

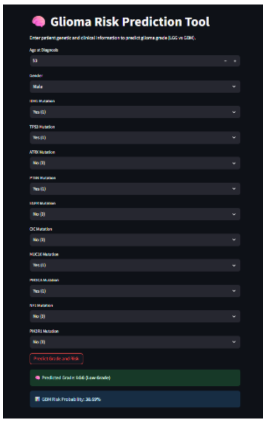

Fig 14. Streamlit Clinical Decision Support Interface

Discussion

The combined causal- machine learning method illustrated strong execution and comprehensibility compared to previous Machine learning only glioma labelers. In contrast to black-box models, the integration of CausalForestDML and SHAP offered biologically based explanations of how genomic interactions built disorder growth. Additionally, discovering IDH1 and ATRX as shielding while TP53, PTEN, and EGFR drive aggressiveness, corresponds with microscopic pathological discoveries in the 2021 WHO CNS Classification. This combination of data-driven inference with field biology determines a reproducible model for precision neuro-onocology.

Conclusion

This study demonstrates the integration of causal inference and explainable machine learning for precise glioma grade prediction using genomic and clinical data. The logistic regression model achieved the best performance (Accuracy = 85.9%, AUC = 0.92), supported by Random Forest cross-validation results (Accuracy = 0.83 ± 0.03, AUC = 0.90 ± 0.01). Causal analysis identified TP53 and PTEN as malignancy accelerators, while IDH1 and ATRX acted as protective biomarkers. SHAP interpretations and counterfactual simulations provided transparent insights into mutation-driven progression, enhancing model trustworthiness. Finally, a Streamlit-based Clinical Decision Support System enables real-time, interpretable glioma risk assessment, bridging computational genomics and practical oncology. This framework highlights the potential of causal explainable AI to advance precision diagnostics and individualized treatment planning in neuro-oncology.

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